

Grid cell move problem

February 14, 2025

1 Statement

You are given a grid and two sets of cells in it, R and G , which denote cells coloured red and green, respectively. Determine the assignments that minimise the total cost, following these rules:

- Assignment types
 - Individual assignment
 - * A single red cell can be directly matched to a single green cell.
 - * The cost of such assignment is the Euclidean distance between the two cells.
 - Group assignment
 - * A *red group* is a pair of two adjacent by edge or corner red cells.
 - * A *green group* is a pair of two adjacent by edge or corner green cells.
 - * A red group can be matched to a green group, effectively pairing both reds and both greens.
 - * The cost of such assignment is the Euclidean distance between the midpoints of the two groups.
- Constraints:
 - If $|R| \leq |G|$, all red cells must be assigned to a green cell.
 - If $|R| > |G|$, all green cells must have an assigned red cell to them.
 - A cell can only be assigned via one single assignment or one group assignment.
 - A coloured cell can be either red or green, not both.

2 An ILP solution

I have modeled and successfully implemented an ILP solution that handles groups. Below is a mathematical statement of the problem.

2.1 Variables

- $x_{i,j} \in \{0,1\}$: Binary variable for individual assignment, where $x_{i,j} = 1$ if red point R_i is assigned to green point G_j .
- $y_{k,l} \in \{0,1\}$: Binary variable for group assignment, where $y_{k,l} = 1$ if red group k (composed of cells R_{k_1} and R_{k_2}) is assigned to green group l (composed of cells G_{l_1} and G_{l_2}).

2.2 Objective Function

$$\text{Minimize } \sum_{i,j} d(R_i, G_j) \cdot x_{i,j} + \sum_{k,l} (d'(k, l)) \cdot y_{k,l}$$

where:

- $d(R_i, G_j)$: Distance between red point R_i and green point G_j .
- $d'(k, l)$: Distance between centroids of red group k and green group l .

2.3 Constraints

2.3.1 Red Constraints (R)

For each red point R_i :

- If $|R| \leq |G|$:

$$\sum_j x_{i,j} + \sum_{\substack{k \ni i \\ l}} y_{k,l} = 1 \quad (\text{Exact assignment})$$

- If $|R| > |G|$:

$$\sum_j x_{i,j} + \sum_{\substack{k \ni i \\ l}} y_{k,l} \leq 1 \quad (\text{At most one assignment})$$

2.3.2 Green Constraints (G)

For each green point G_j :

- If $|R| > |G|$:

$$\sum_i x_{i,j} + \sum_{\substack{l \ni j \\ k}} y_{k,l} = 1 \quad (\text{Exact assignment})$$

- If $|R| \leq |G|$:

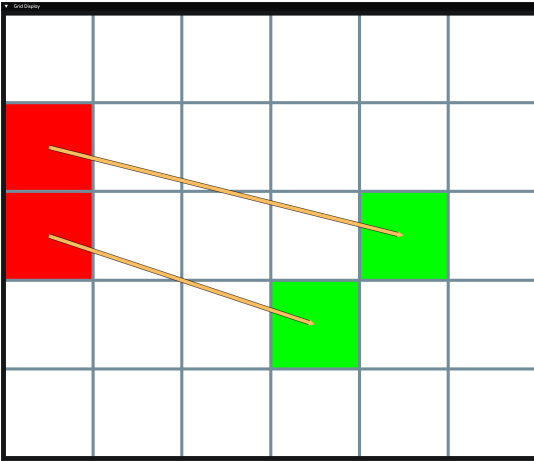
$$\sum_i x_{i,j} + \sum_{\substack{l \ni j \\ k}} y_{k,l} \leq 1 \quad (\text{At most one assignment})$$

3 Other possible solutions

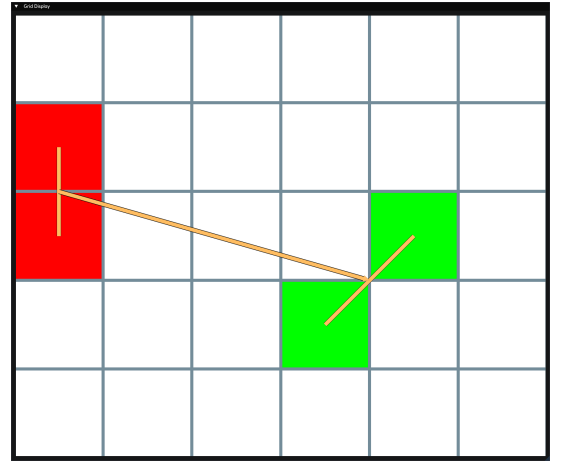
Other possible solutions can utilise similar problems, such as variations of the assignment problem class as LSAP.

Greedy approaches can prioritise group assignments via heuristics and solve the remaining cells using the Hungarian Algorithm (for LSAP(Least-Sum Assignment Problem)).

4 Examples

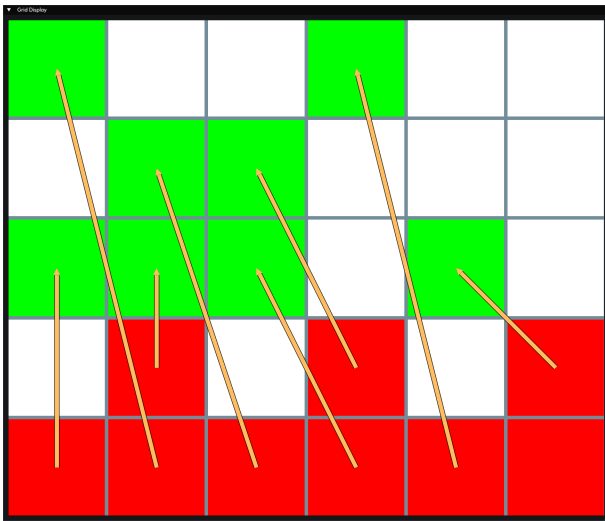


(a) Optimal individual assignment

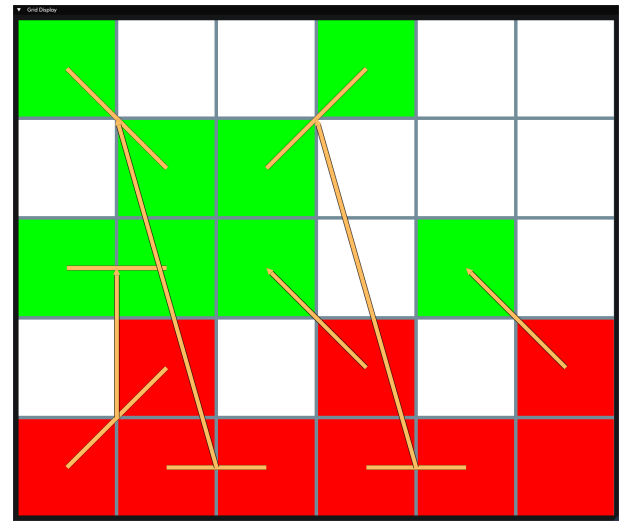


(b) Optimal group assignment

Figure 1: Example 1

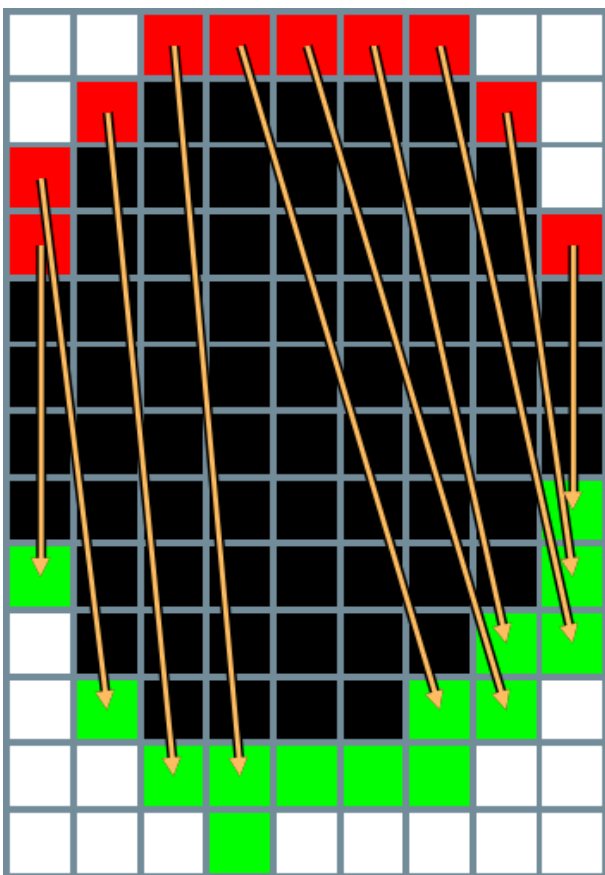


(a) Optimal individual assignment

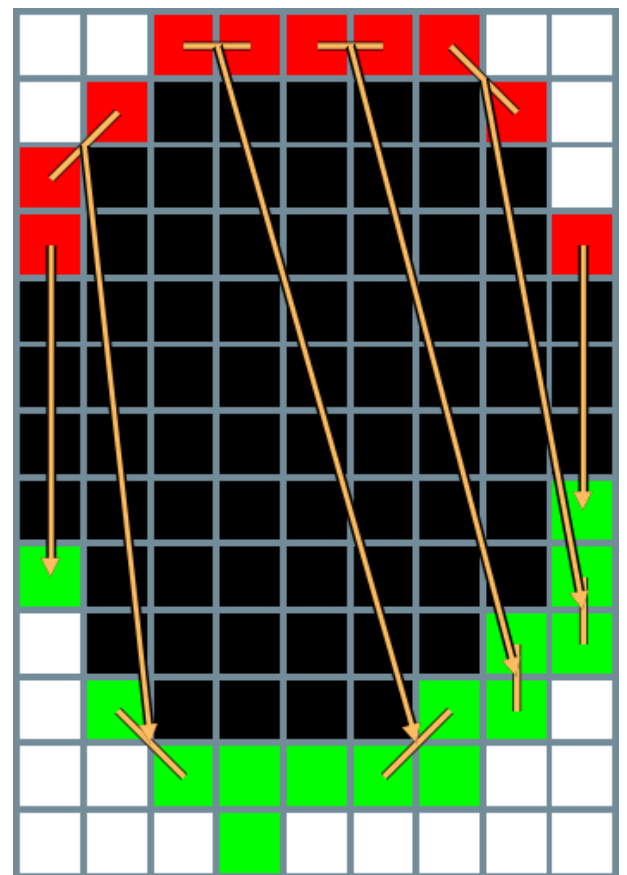


(b) Optimal group assignment

Figure 2: Example 2



(a) Optimal individual assignment



(b) Optimal group assignment

Figure 3: Example 3